Section 1: Multiple Choice (Choose the correct answer)

1. What happens to electrons in a covalent bond?

   a) They are transferred from one atom to another
   b) They are shared between atoms
   c) They are lost completely
   d) They disappear

2. Which of the following elements is most likely to form a covalent bond?

   a) Sodium (Na)
   b) Magnesium (Mg)
   c) Carbon (C)
   d) Potassium (K)

3. Which of the following molecules contains a double bond?

   a) H_2O
   b) O_2
   c) NH_3
   d) CH_4

4. What is the name of the covalent compound CO_2?

   a) Carbon dioxide
   b) Carbon monoxide
   c) Carbon oxide
   d) Dicarbon monoxide

5. Which bond is the most polar?

   a) C-H
   b) O-H
   c) N-H
   d) F-F

6. What is the formula for nitrogen (II) oxide?

   a) NO
   b) NO_2
   c) N_2O
   d) N_2O_4

7. Which formula represents the structure of a saturated hydrocarbon?

a) C_2H_2 b) C_4H_{10} c) C_5H_8 d) C_6H_6

Section 2: Short Answer 6. Explain the difference between an ionic bond and a covalent bond.

7. Draw the Lewis dot structure for ammonia (NH_3).

8. What type of bond forms between nitrogen (N) and hydrogen (H), and how many bonds does nitrogen typically form?

9. Draw a dot and cross diagram for the ions Mg and N.

10. Write the correct name for the following covalent compounds:

a) PCl_3
b) N_2O_4
c) SF_6

11. Write the correct name for the following ionic compounds:
a) FeO

b) Fe_2O_3

c) $PbSO_4$

d) SnS_2

Section 3: Diagram-Based Questions

11. **Explain** whether the following structural formulas represent covalent or ionic compounds:

a) NaCl

b) CH_4

c) CO

d) MgO

12. Label the partial charges (δ+ and δ−) on a water (H_2O) molecule and explain why water is a polar molecule.

Section 4: Application

13. Max and Jumpei are working on a science project. Max wants to create a compound that will conduct electricity in water, while Jumpei wants a compound that shares electrons equally between atoms. What type of bond should each person look for?

14. A spaceship is exploring a distant planet and discovers an unknown element. The scientist Giovanna noticed that it readily forms compounds with 2 fluorines by sharing electrons. Based on this, what type of bond is likely forming, and how many bonds can the unknown element form?

15. Sophia is baking cookies and realizes that table salt (NaCl) dissolves in water, but sugar ($C_{12}H_{22}O_{11}$) does not conduct electricity when dissolved. Explain why this happens in terms of bonding and structure.

16. In a chemistry lab, Zoe and Amit are testing the properties of different substances. One substance has a high melting point and conducts electricity when dissolved in water, while another has a low melting point and does not conduct electricity. What can you infer about the types of bonds in these substances?

17. Describe the concept of allotropes and provide an example of an element with multiple allotropes.

18. Draw as many of the 5 isomers of C_6H_{14}.